## *Is Capability Repair Competitive in Small Language Models?*

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### 1. Path Choice

Engineer Track; Option 1: TinyStories LLM; Individual.

### 2. Research Question and Engineering Goal

- **Background:** Small language models often fail in specific ways, such as repetition, broken entity tracking, or logical inconsistency. A common solution is to create a small "patch" dataset and fine-tune the model to fix a particular failure. However, it is unclear whether this repair is local or competitive. Large models, due to their abundant parameter capacity, typically do not suffer from severe trade-offs when improving one metric. However, it remains unclear whether a TinyStories-scale model has sufficient capacity to avoid such competition.
- **Goal:** In this project, I aim to determine whether improving one failure type in a TinyStories model can be achieved without harming other abilities. By quantifying the trade-offs between different capabilities, this project seeks to understand the capacity limits of small models better and provide practical guidance for stable capability enhancement in low-parameter LLMs.

### 3. Methodology

- Base Model and Dataset: Use a language model trained on TinyStories, and generate outputs on a diverse evaluation set.
- Define Failure Categories: Such as repetition, entity tracking errors, coherence breaks, and factual inconsistencies.
- Build automatic or semi-automatic metrics to measure each category.
- Patching: Construct a small patch dataset that focuses only on correcting one failure type (one patch for each type if necessary), then fine-tune the model using strategies such as full fine-tuning and LoRA fine-tuning.
- Evaluation: By measuring the improvement in the target failure and the change in the non-target abilities, we can compare metrics like target improvement, non-target degradation (interpreted as catastrophic forgetting), and the difference between full fine-tuning and LoRA. Finally, the project will determine whether capability repair in small language models is competitive and whether techniques such as LoRA can alleviate catastrophic forgetting.

### 4. Related Work

- *TinyStories: How Small Can Language Models Be and Still Speak Coherent English?* (Eldan & Li, 2023)
  It shows that model size alone does not determine capability, and dataset design significantly influences small-model performance, which provides a foundation for evaluating whether small language models can remain competitive under different fine-tuning strategies.
- *LoRA: Low-Rank Adaptation of Large Language Models* (Hu et al., 2021)
  LoRA proposes a parameter-efficient fine-tuning method that injects trainable low-rank matrices into transformer layers while freezing the base model weights. It significantly reduces trainable parameters while maintaining competitive downstream performance.
- *Analyzing and Reducing Catastrophic Forgetting in Parameter Efficient Tuning* (Ren et al., 2024)
  It proposes a method, I-LoRA, which combines LoRA parameter interpolation with experience replay to improve the stability of large language models under continual fine-tuning, and shows that compared to standard PEFT methods, I-LoRA significantly reduces forgetting. Although the study mainly focuses on large models, its approach provides useful insights into understanding the mechanisms of forgetting under parameter-efficient fine-tuning.